Strike Pi Core Studio 2.0.16 User Guide

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This guide explains every module, control and firmware feature in the project: what it does, how to configure it, its defaults and ranges, how the firmware behaves, and the limits you should know about.

* Read this first: PS4 authentication requires key material extracted from hardware you own. This project ships no Sony private keys and has no fallback key.

The app is a management console, not a runtime component. Once settings are saved to the Pico, you can close the app and play; the Pico keeps working.

Quick start

Flash the firmware

1. Unplug the Pico 2W.

2. Hold **BOOTSEL** while plugging it back in.

3. A removable volume appears, usually RPI-RP2 (the app also accepts any removable volume that contains NFO_UF2.TXT).

4. In the app, open **Firmware Tools**, choose the uf2, press Validate & hash, then Copy UF2 to Pico You can also copy the file to the volume manually.

5. The Pico reboots automatically. On first boot it presents the Sony DualShock 4 identity.

6. The Pico Need Bond So go to Firmware Tools Tab And click The **Bond Pico** gives this specific bridge its own unique Bluetooth device address and controller serial number. Make sure the controller is connected.

Pair the DualShock 4 Or Dual Sense Controller

1. Power the Pico from a USB data port.

2. Put the DS4 in pairing mode: hold **Share + PS** until the light bar flashes rapidly (the exact gesture depends on the controller revision).

3. The Pico scans automatically. The on-board LED stays Flashing while no controller is connected and stays ON while a controller is connected.

4. The controller connects and the Pico starts forwarding input.

If a different controller was paired before, use Combo Buttons. Or open Configuration Connection and press Disconnect controller. The firmware drops the link and deletes the stored Bluetooth link keys, so the next controller can pair. Pairing starts again automatically.

Connect the app

1. Launch Strike Pi Core Studio-BETA.exe.

2. The app probes USB automatically on startup. You can also press **Connect Pico** in the sidebar. Mod Mode **must be ON** (the configurator report must be present). A DualShock 4 must be **paired and streaming** (Bluetooth linked). Use Combo Buttons to Turn This Mode ON.

3. A successful connection shows the device name, `VID:PID`, firmware version, protocol revision, identity and auth state.

If the app reports that the firmware does not support the v2 management Commands, the Pico is running an older build. Flash firmware 1.0.17.

PC gameplay Or other Consoles that support It.

1. Open **Identity**.

2. Select **Sony DualShock 4** and press **Apply selected identity**. Or use Combo Buttons.

3. Wait for the USB re-enumeration and automatic reconnect.

4. Windows, Steam Input and Windows or other supported consoles see a normal DualShock 4.

No PS4 keys are needed for PC or supported Consoles play.

PS4 gameplay

1. Open **Authentication** and import:

   - key.pem a complete RSA-2048 private key,
   - serial.txt the 16-hex-character DS4 serial,
   - sig.bin the 256-byte Sony certificate signature.

2. Press **Validate files**. All local checks must pass.

3. Press **Upload to Pico**. The Pico writes the record, re-verifies it and reports success.

4. Open **Identity**, select **Razer Panthera**, and apply it. Or use Combo Buttons.

5. Connect the Pico's USB output to the PS4.

The PS4 runs its F0 F3 challenge against the stored key. Authentication is local to the Pico; the app does not need to be running.

A PS4 does not accept this bridge through the Sony DS4 identity. Use Panthera for the console and switch back to Sony for PC play or other supported consoles. Windows may enumerate the Panthera, but its button layout is that of an arcade stick.

PS3 gameplay

1. Open **Identity**.

2. Select **Sony PS3 Controller** and press **Apply selected identity**. Or use Combo Buttons.

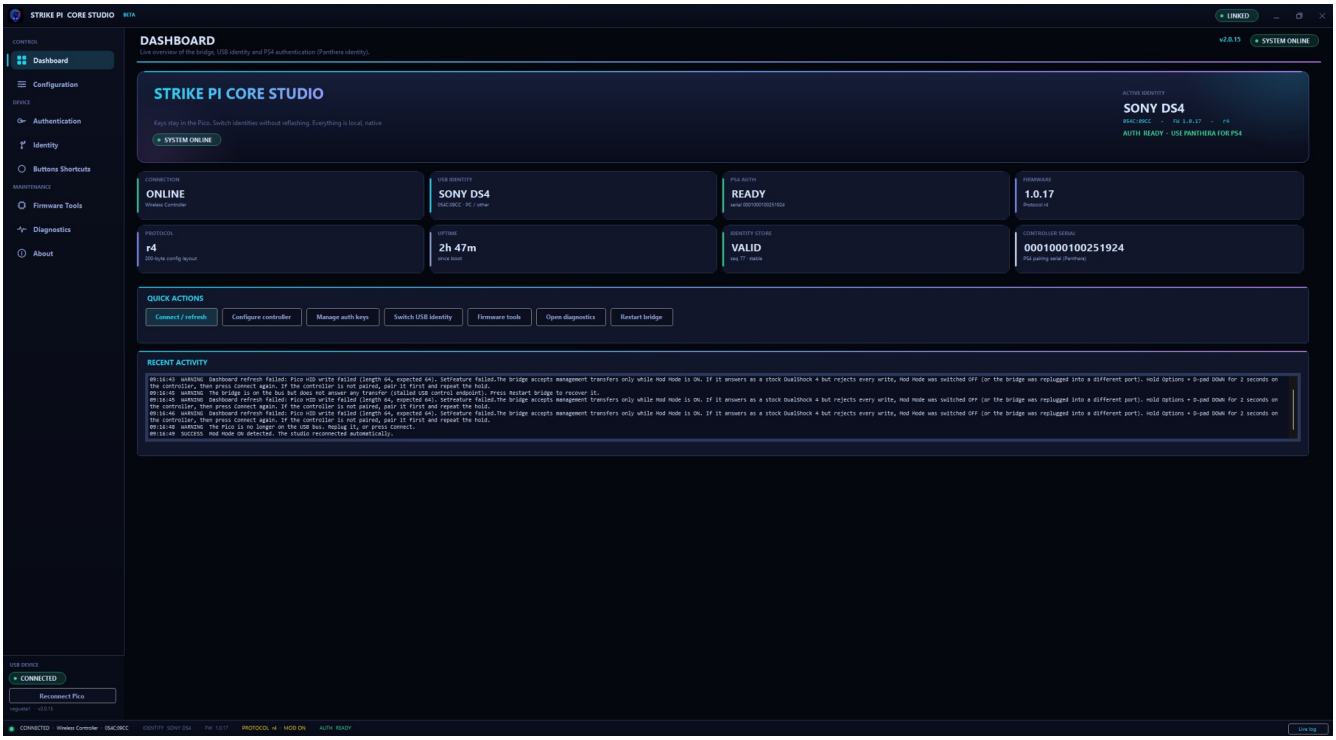
3. Wait for the USB re-enumeration and automatic reconnect.

4. PS3 console see a normal DualShock 3.

No PS4 keys are needed for PS3 play.

Dashboard

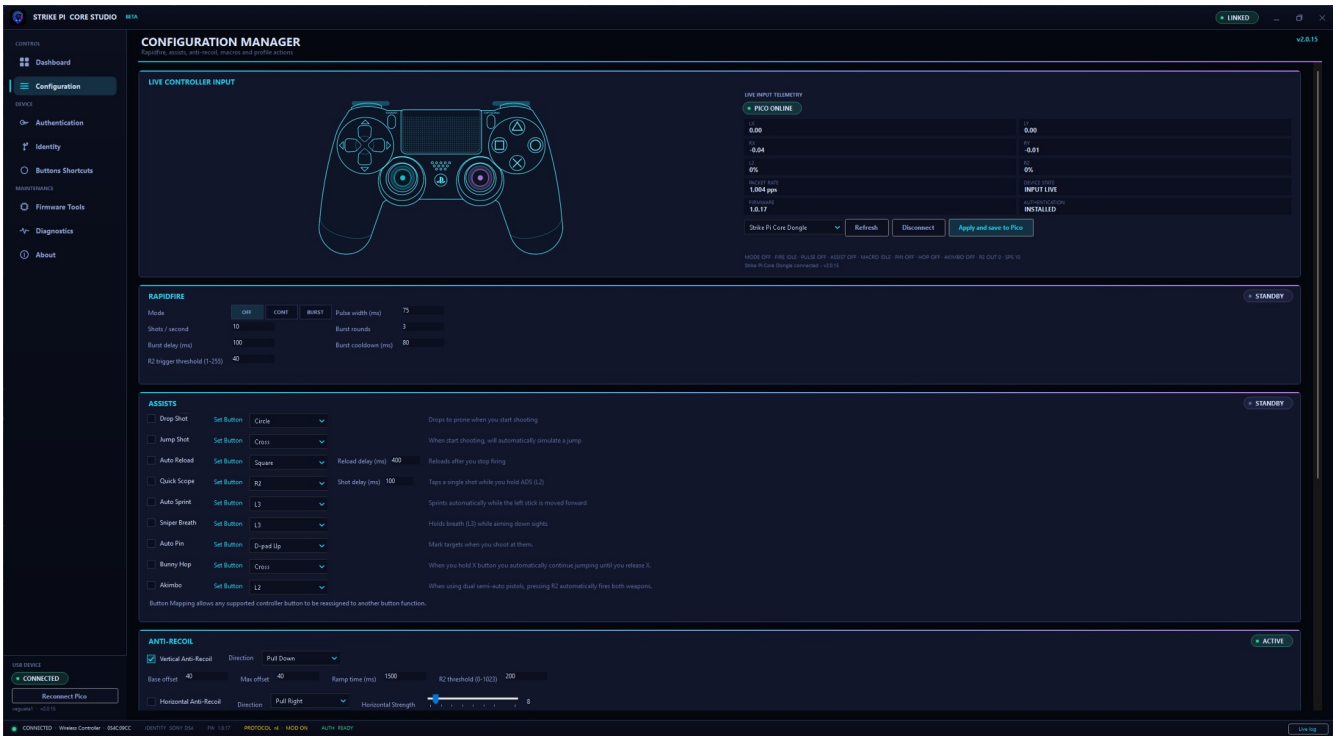
The Dashboard is a read-only overview with shortcuts.

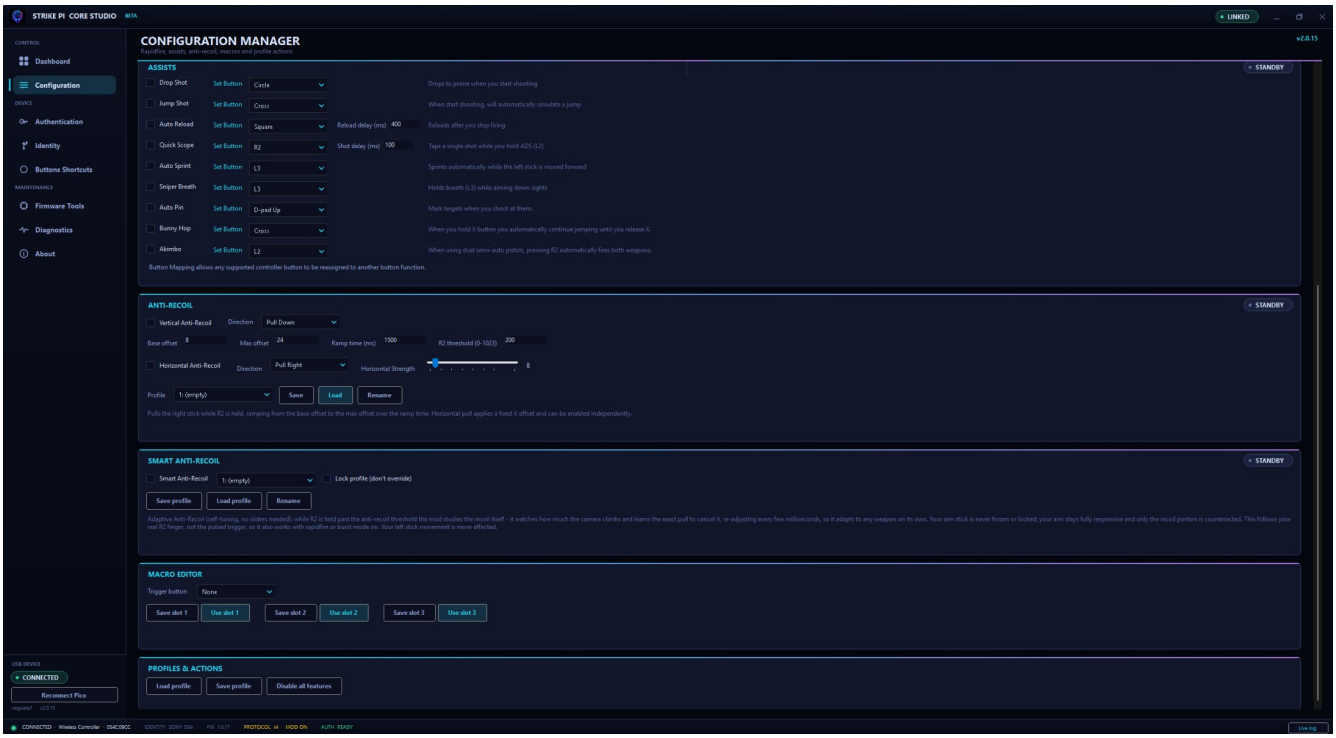


Quick actions

Connect / refresh Connects if offline, then refreshes the cards
Configure controller Opens Configuration
Manage auth keys Opens Authentication
Switch USB identity Opens Identity
Firmware tools Opens Firmware Tools
Restart bridge Restarts the Pico management link (firmware 1.0.17+)

Configuration Manager





The Configuration page is the main feature editor. It is organized into seven cards:

1. Live Controller Input
2. Rapid fire
3. Assists
4. Anti-Recoil
5. Smart Anti-Recoil
6. Macro Editor
7. Profiles & Actions

Practical notes

- Rapid fire works best with semi-automatic weapons. Fully automatic weapons already generate their own fire rate and may ignore or fight synthetic trigger pulses.

- Higher SPS is not automatically better: many games poll input at a fixed rate And cap how fast a weapon can fire. Tune SPS and in an offline test.

- The real R2 value is sampled before rapid fire rewrites the output. Assists, anti-recoil and smart anti-recoil therefore follow your finger, not the synthetic pulse train.

Assists card

Each mapped assist has an enable toggle, a selectable output button, and a one-line description. The activation inputs remain fixed. All assists are off by default.

Drop Shot Circle Presses the mapped button while you fire, with a stand-up tap sequence on release.

Jump Shot Spams the mapped button while R2 is held.

*Auto Reload*Square Taps reload after you stop firing Configurable 0-2000 ms delay (400 ms default).

Quick Scope R2 When ADS starts (L2), waits the shot delay and taps one shot Configurable 0-2000 ms delay (100 ms default).

Auto Sprint L3 Spams the mapped button while the left stick is pushed forward.

Sniper Breath L3 Holds the mapped button while L2 is held Continuous while ADS.

Auto Pin D-Pad Up Sends two D-Pad Up taps on the physical R2 rising edge.

Bunny Hop Cross Repeats the mapped button only while Cross is physically held.

Akimbo L2 mirrors R2 Copies the final R2 output to L2, preserving any stronger physical L2 input, works with Rapid-fire Continues mode and burst mode too.

Anti-Recoil card (vertical + horizontal)

This card controls the deterministic, manually tuned anti-recoil.

- Vertical Anti-Recoil- Enables the Y-axis pull. Direction of the correction Pull Down / Pull Up.

Base offset Initial pull in strength.

Max offset Pull strength reached at the end of the ramp.

Ramp time (ms) Time to interpolate from base to max.

R2 threshold Trigger level that activates anti-recoil.

- Horizontal Anti-Recoil- Enables the X-axis pull. Direction of the correction Pull Right / Pull Left.

Horizontal Strength Fixed X offset in Pull strength.

Anti-recoil profile slots (5)

- Stored in `antirecoil-profiles.json` beside the executable.

- Each slot snapshots: enabled, direction, base, max, ramp, threshold, X enabled, X direction and X strength.

- *Save* writes the current controls into the selected slot.

- *Load* restores the slot into the controls. It does not talk to the Pico.

- *Rename* opens a small dark dialog. Renaming an empty slot seeds it with the current settings so a name never exists without a matching snapshot.

- Loading an empty slot shows empty use Save first.

- Corrupt slot files fall back to empty slots instead of crashing.

Smart Anti-Recoil card

Smart Anti-Recoil (also called Adaptive Anti-Recoil, AAR) is a self-tuning loop with no sliders. It learns the recoil from how you compensate and adds a correction on top of your live aim.

How it works

- Activation uses the same physical-R2 threshold as anti-recoil. It works with rapid fire because the trigger snapshot is taken before rapid fire rewrites the output.

- Every 10 ms it measures the aim stick's offset from center on both axes.

- The residual is re-signed using:

Y uses the *vertical anti-recoil direction* (Pull Down / Pull Up).

X uses the *horizontal anti-recoil direction* (Pull Right / Pull Left).

- Residuals at or below 6 stick units count as dead-zone jitter and are not learned.

- The learned pull is integrated in fixed point with a gain of about 4 units per residual unit per tick. Strong sustained compensation converges faster; tiny residuals drift slowly.

- The learned pull is clamped to 100 stick units. It only grows in the chosen pull direction; over-correction (camera dipping) reduces it.

- The learned value is added *on top of your live stick*, not instead of it. Aiming is never frozen or locked.

- The left stick (movement) is never touched.

- The learned value persists between bursts and across configuration saves while the controller stays connected. It resets on a controller link timeout, engine reset or reboot.

- It adapts automatically when you switch to a weapon with a different recoil impulse.

- You can activate Smart-Anti Recoil on top of Vertical Anti-Recoil to get the best compensation you can get. I use Vertical Anti-Recoil 40 and 40 max with this and works with any weapon on the game without tuning nothing.

Macro Editor card

The macro editor plays a sequence of button presses. It is a single-pass macro:
one run per trigger press, up to 20 steps.

Trigger button Button that starts the macro.

Save slot 1 to 3 Stores the current trigger + steps in a local slot.

Use slot 1 to 3 Loads the slot and applies immediately if connected.

+ Add step Up to 20 Adds a step row.

Button Button pressed by the step.

Hold ms 0 to 5000 Press duration.

Hold until macro ends On / Off Latches the button until the macro finishes.

Delay ms 0 to 5000 Wait after the step before the next step.

Red X symbol Removes the step.

Profiles & Actions card

Load profile Loads a .json profile into the app controls.

Save profile Saves all current controls and macro steps to a .json file.

Disable all features Reads the Pico config, turns every feature flag off and writes it back.

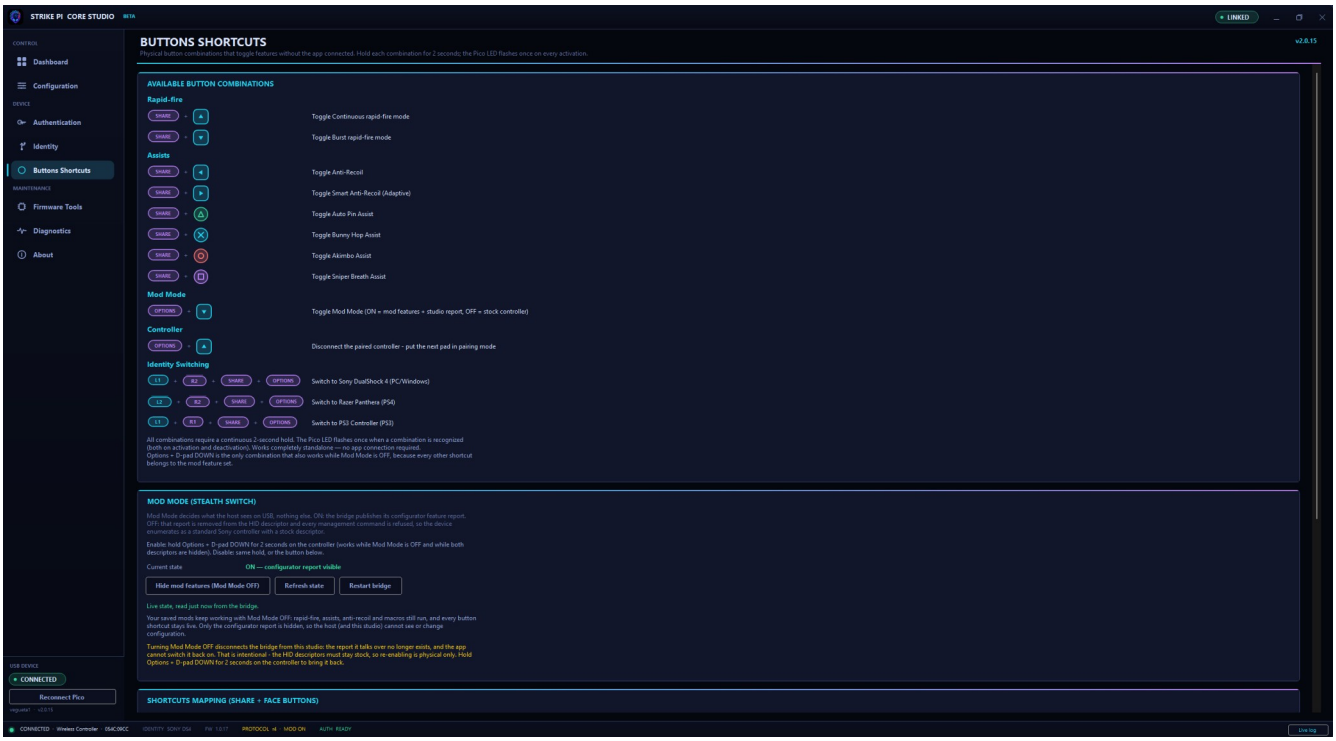
What Disable all features clears:

- Rapid-fire mode Off.
- Drop Shot, Jump Shot, Auto Reload.
- Quick Scope, Auto Sprint, Sniper Breath.
- Macro enable and macro step count.
- Vertical anti-recoil, horizontal anti-recoil, smart anti-recoil.

It leaves button maps, tuning values and profile files intact. It then reloads
the controls from the Pico so the UI shows the disabled state.

Button Shortcuts

The same settings can be toggled from the controller without the app. Every combination needs a *continuous 2-second hold*; the Pico LED (on the wireless module) flashes once when the hold is recognized, both when turning a feature on and when turning it off. A quick tap does nothing.
Mod Mode (stealth switch) Mod Mode decides what the host sees on USB. Need to be On when you are using the App and need to be OFF when You playing is a Protection layer. Rapid fire, assists, anti-recoil, macros and all button shortcuts continue running when Mod Mode is OFF.



Share + D-pad UP Toggle Continuous rapid-fire mode
Share + D-pad DOWN Toggle Burst rapid-fire mode
Share + D-pad LEFT Toggle Anti-Recoil
Share + D-pad RIGHT Toggle Smart Anti-Recoil (Adaptive)
Share + Triangle Toggle Auto Pin (or the remapped assist)
Share + Cross Toggle Bunny Hop (or the remapped assist)
Share + Circle Toggle Akimbo (or the remapped assist)
Share + Square Toggle Sniper Breath (or the remapped assist)
L1 + R2 + Share + Options Switch to Sony DualShock 4 (PC/Windows)
L2 + R2 + Share + Options Switch to Razer Panthera (PS4)
L1 + R1 + Share + Options Switch to PS3 Controller (PS3)
Options + D-pad UP Disconnect the paired controller and return to pairing mode
Options + D-pad DOWN Toggle *Mod Mode* (ON = mod features + studio report, OFF = stock controller)

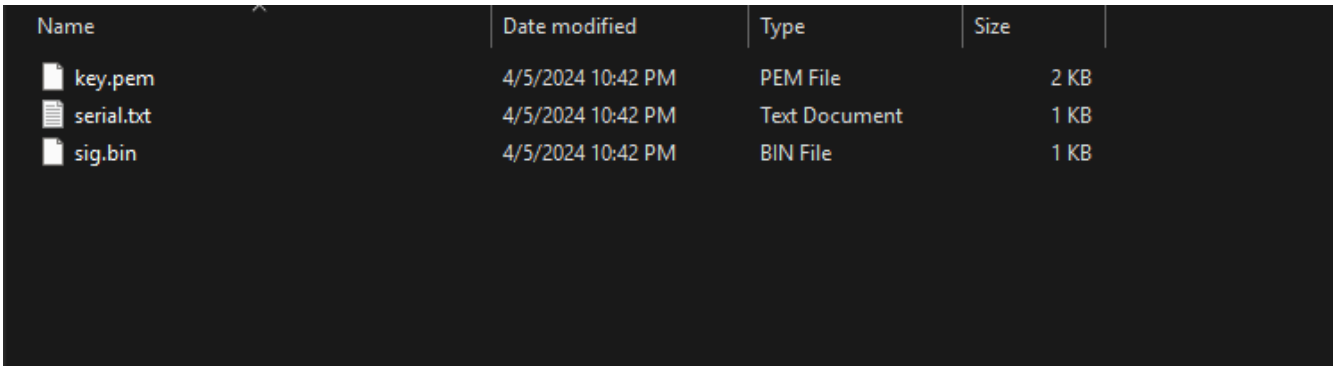


You can Map any assists on Shortcuts Mapping replacing the defaults ones.

Authentication Manager

This page manages the PS4 key material stored in the Pico's flash sector. The app never ships a key and the firmware never falls back to a built-in key.



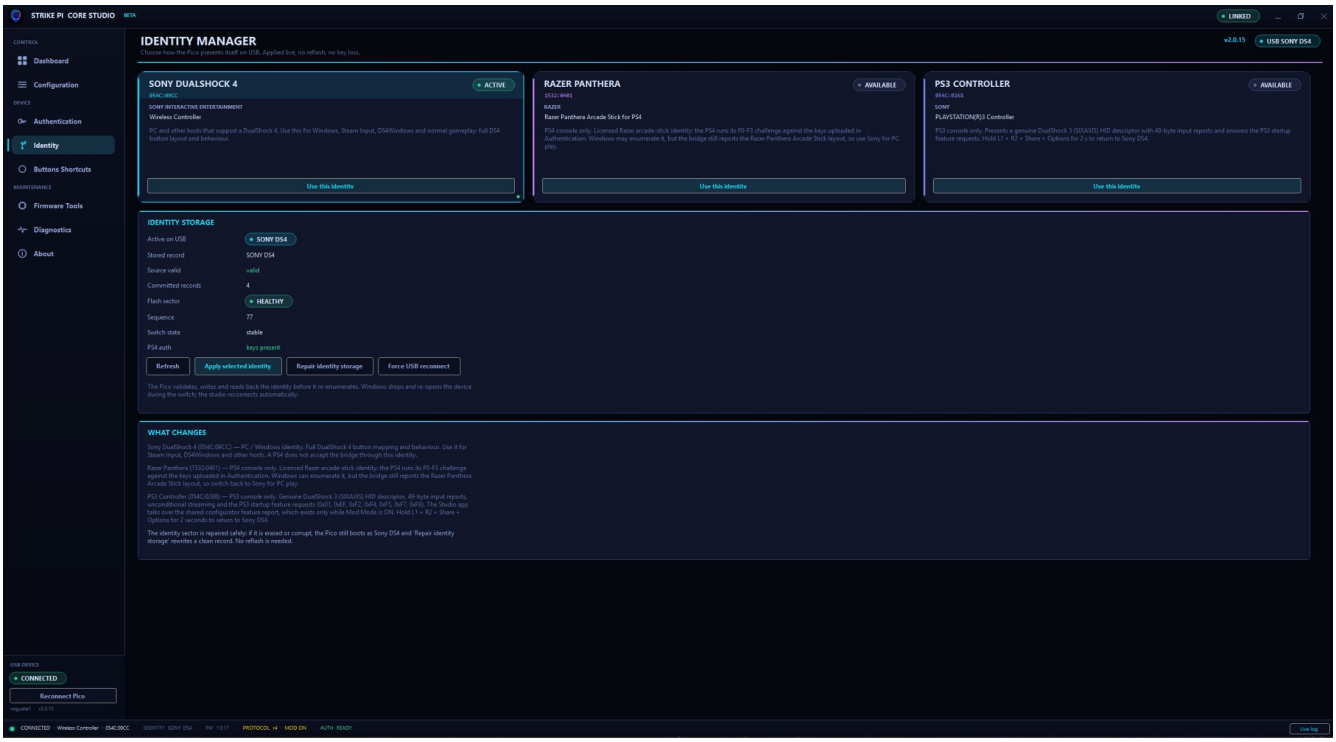


What the three files are Required content:

key.pem – A complete RSA-2048 private key in PEM format.
serial.txt- The 16-hex-character DS4 serial, or a 32-hex-character serial field.
sig.bin- Exactly 256 bytes: the Sony certificate signature.

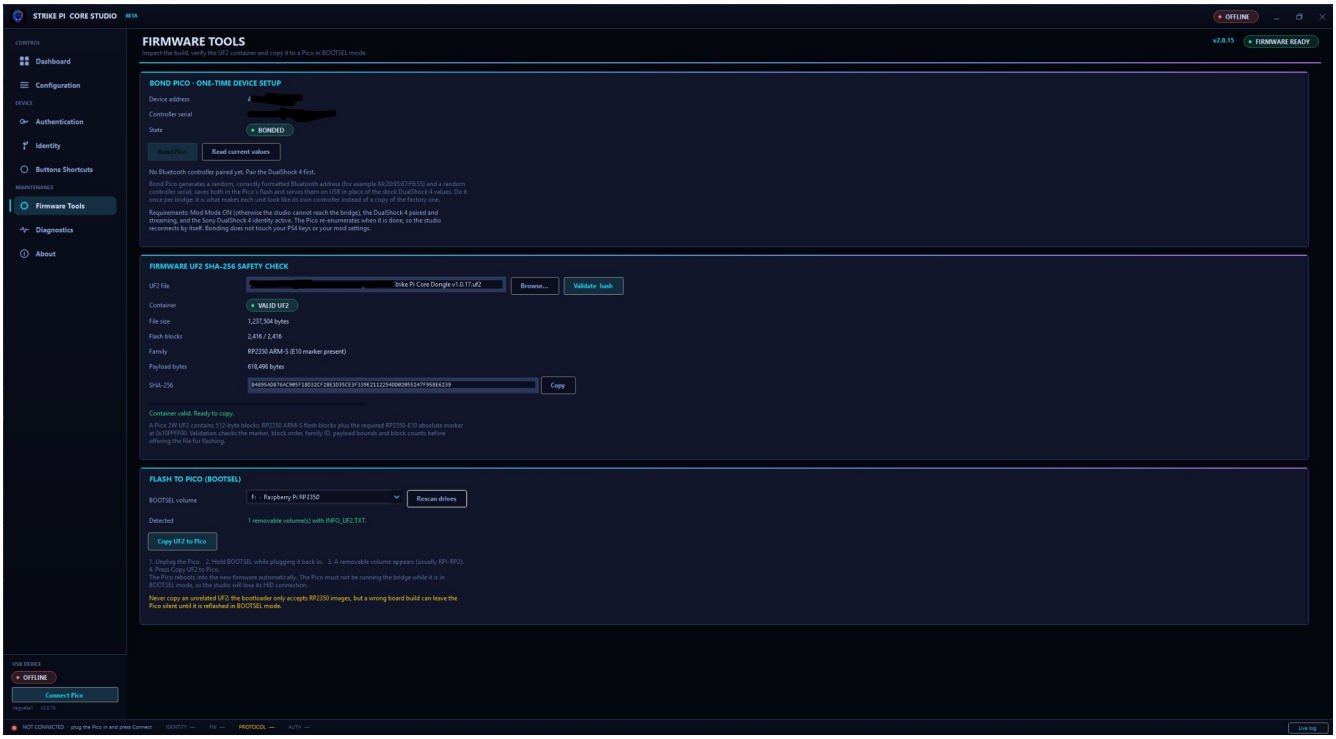
Identity Manager

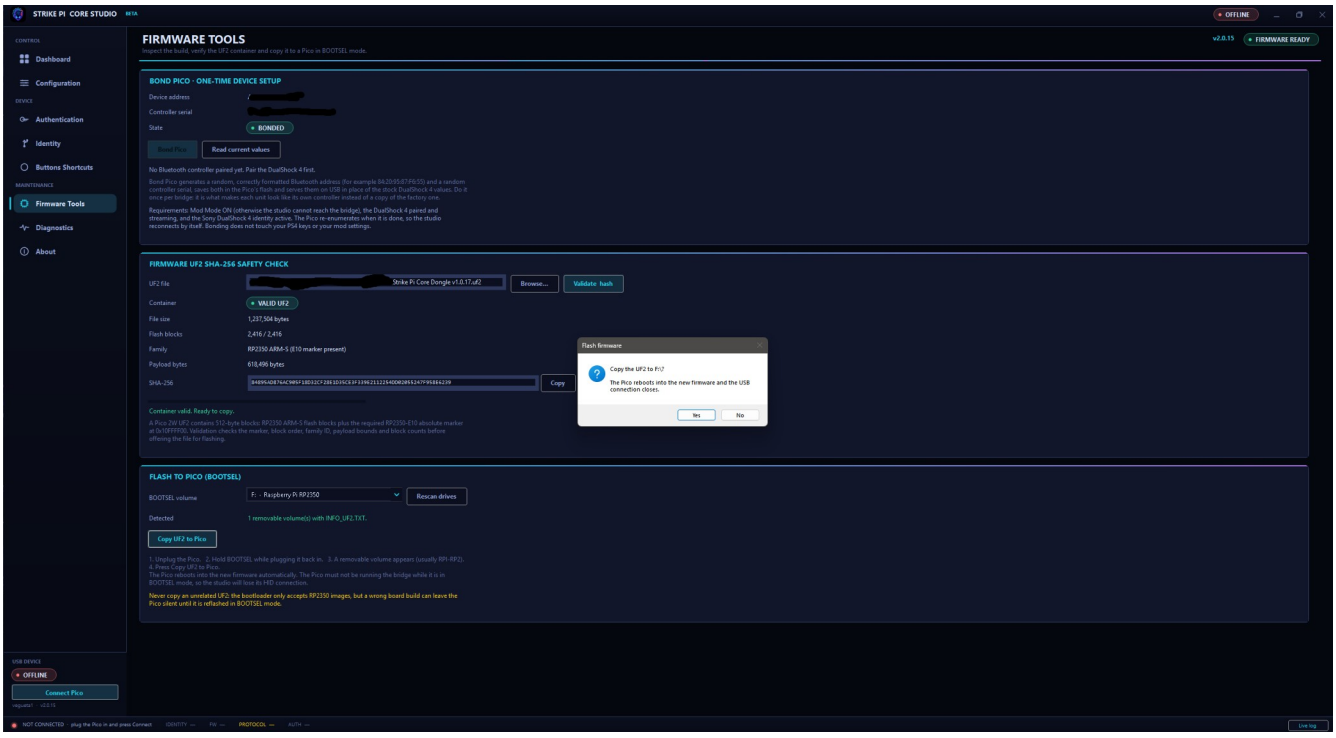
The Identity page switches the Pico's USB personality and reports the health of the identity store.



Firmware Tools

Firmware Tools validates a UF2 container and copies it to a Pico in BOOTSEL mode. It does not compile firmware. The Pico cannot run the bridge while it is in BOOTSEL mode, so the app will lose its HID connection during flashing. That is expected.





Bond Pico card

The *Bond Pico* card gives this specific bridge its own unique Bluetooth device address and controller serial number. Every unit ships with the same stock values (Bluetooth 01:02:03:04:05:06, serial 0001000100251924), so a host-side dashboard or OS sees every bridge as the identical controller and reports *N/A* for the USB serial. Bond Pico fixes this once per device.

About

The About page documents the build and the executable itself.



The hash is computed the first time the page becomes visible. Compare it with the published checksum in `SHA256SUMS.txt` or the release notes to verify the file you received.

Troubleshooting

The app cannot find the Pico:

- Use a **data** USB cable, not a charge-only cable.

- Confirm the Pico is running the bridge firmware, not BOOTSEL mode.

- Try another USB port; avoid unpowered hubs.

- Close or pause DS4Windows, Steam Input, VorpX, overlays and other software that may hold the HID interface.

- Press **Connect Pico** again.

- In Device Manager, confirm a device with `054C:09CC`, `1532:0401` or `054C:0268` exists.

- If the Pico is in BOOTSEL mode, it exposes a mass-storage volume, not the HID management interface.

- **PS3 mode only:** a third-party DualShock 3 driver that claims `USB\VID_054C&PID_0268` (DsHidMini installs itself for exactly that hardware ID) takes the device before Windows' HID driver. The device appears in Device Manager as *DS3 Compatible HID Device* and no HID interface exists, so the app cannot open it. The studio detects this and names the driver. Either unbind it for this device (Device Manager > right-click > Update driver > Browse > Let me pick from a list > **USB Input Device**) or keep the bridge on the Sony DS4 identity. After unbinding, a USB node restart may be needed before the HID collection appears.

Smart anti-recoil does not adapt:

- Smart learns from **your compensation**, so you must fire in a controlled burst and let the camera climb; it is not a fixed value.

- It only learns while R2 is past the threshold.

- Direction comes from the vertical/horizontal direction controls even if those features are off.

- If you loaded a smart profile, **Lock profile** may be overriding the checkbox. Uncheck it to change the enable state.

- The learner resets on a controller link drop, engine reset or reboot. It does not persist across power cycles.

- If it over-corrects, the learner should reduce the pull; if it still feels wrong, switch direction and re-test.

Safety and legal

- **Use your own hardware and keys.** The project does not supply Sony key material. Importing keys extracted from hardware you do not own may violate law or contracts.

- **Keys are sensitive.** Anyone with the private key, serial and signature can impersonate the console pairing. Store backups encrypted and offline.

- **No warranty.** The software is provided as-is. A wrong firmware image, a power loss during flashing or a corrupted configuration can leave the device temporarily unusable until re flashed.

- **Not affiliated with Sony or Razer.** Sony, PlayStation, DualShock and Razer are trademarks of their respective owners.

- **Do not use this project to bypass console protections or to cheat in competitive or online environments.** The author does not condone that use.